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The empirical evidence of audience costs theory: a quasi-meta-analysis of case evidence based on Bayesian logic

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Abstract

James Fearon's "Domestic Political Audiences and the Escalation of International Disputes" is arguably one of the most influential articles in International Relations. This article takes stock of the enterprise of audience costs theory (ACT) by providing a "quasi-meta-analysis" of the case studies (QMACS) for ACT based on an explicit Bayesian logic. I show that overall evidence for ACT has been quite thin, whereas evidence against it has been quite powerful. While audience costs do exist, they are much more contingent, and their magnitude is much smaller than what ACT's proponents have believed. Because audience costs are only one possible pathway that links domestic politics with states' international behavior, and domestic politics is far messier than proponents of ACT are willing to admit, overselling audience costs is unwarranted. To better understand the relationships between domestic politics and international behaviors, we need to move beyond audience costs and bring in other mechanisms. By synthesizing different case studies scattered in the literature with an explicit Bayesian logic, QMACS provides a new tool for disciplining and synthesizing case studies regarding other prominent theories such as democratic peace, extended deterrence, inequality and democratic transitions, and revolution.

Keywords

Quasi-meta-analysis of case studies (QMACS), Bayesian logic, audience costs, costly signaling, crisis bargaining, domestic politics

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> "Some essential part of the study of mixed motive games is necessarily empirical . . . One cannot, without empirical evidence, deduce what understandings can be perceived in a nonzero-sum game of maneuver no more than one can prove, by purely formal deduction, that a particular joke is bound to be funny."
>
> (Thomas Schelling, *The Strategy of Conflict*, 162, 164)

> "Even if deductive models are parsimonious, rigorous, and produce fascinating propositions, they will remain little more than toys unless they can be verified."
>
> (Robert Jervis, 1989: 184)

Introduction

In 1994, James Fearon published arguably one of the most influential articles in International Relations: "Domestic Political Audiences and the Escalation of International Disputes" (Fearon, 1994b; see also Fearon, 1992, 1994a, 1997). This paper formally launched the enterprise of "(domestic) audience costs," defined as "costs that would be suffered if the leader backed down or backed away from a public threat or warning issued in a crisis" (Fearon, 1997: 69; see also Fearon, 1994b: 581; Schultz, 2001a: 33–34; Slantchev, 2012: 377; Tomz, 2007: 821). By 2025, the total citations of Fearon (1994b) have reached 3,864 (Figure 1) and keep growing. In short, the audience costs theory (hereafter, ACT) has become a runaway success. Partly because of ACT, Fearon has been consistently ranked as one of the most influential scholars in International Relations.[1]

As ACT has become a highly influential research enterprise in crisis bargaining, much effort has been devoted to extending, refining, and testing it. The extensive literature that followed Fearon's seminal article thus can be divided into two broad categories: one that critically extends and modifies Fearon's original model (e.g. Schultz, 1998, 2001a; Slantchev, 2011; Tarar and Leventoglu, 2009, 2013) and one that tests key empirical implications produced by Fearon's original model and its extensions. This article focuses on the second body of literature. After all, as Schelling (1960: 164), Jervis (1989: 184), and many others have noted, even with an elegant model, we want to know whether it captures something in the real world. Three full decades should be enough for a stocktaking of ACT's empirical support.

Within the second body of literature, there are three types of tests: qualitative (i.e. case studies, with or without process tracing), quantitative, and experimental. This article focuses on the qualitative type because a case study with process tracing is the most suitable method for revealing whether audience costs (or consideration of them) have been

1. Google Scholar search, 20 April 2026. Unless stated otherwise, audience costs are domestic in the discussion here. The following year, Fearon (1995) published "Rationalist Explanations for War," another highly influential article. On Fearon's stature in the field, see https://trip.wm.edu/data/dashboard/faculty-survey (accessed 22 July 2023).

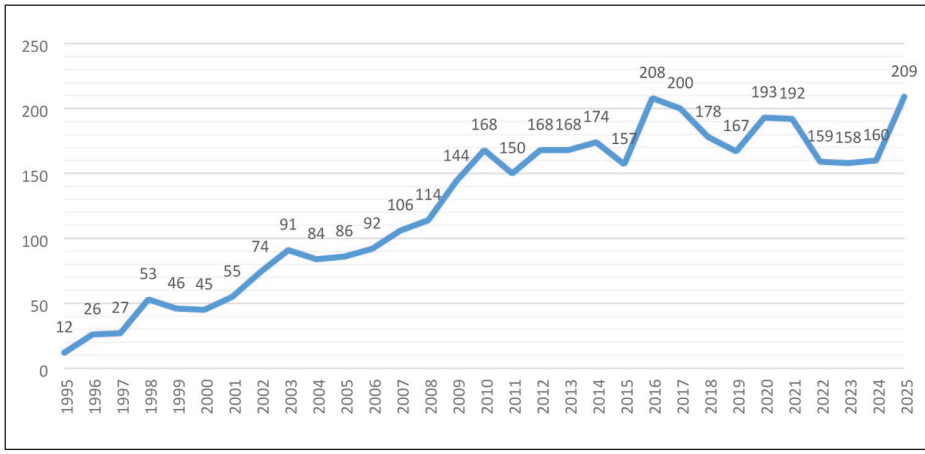


Figure 1. Citations of Fearon (1994b), 1995–2025.

a critical driver in crisis bargaining (Schultz, 2001a, 2001b, 2012).[2] Unsurprisingly, many contributions, most prominently Snyder and Borghard (2011) and Trachtenberg (2012a), have deployed case studies to test ACT.

Unfortunately, these case studies have been scattered in the literature. Moreover, they have not been explicitly structured according to the accepted guidelines of case studies with process tracing advanced by Van Evera (1997), Collier (2011), Mahoney (2012), Bennett (2015), and Beach and Pedersen (2016, 2019), among others. Coupled with the fact that audience costs are not easily quantifiable or even observable (Schultz, 2012), ACT's proponents tend to ignore or even dismiss evidence from case studies as either irrelevant or invalid.

More critically, Snyder and Borghard (2011) and Trachtenberg (2012a) did not explicitly consider the "(self-)selection effects" in crisis bargaining (Fearon, 2002), that is, leaders self-select themselves into either publicly escalating a crisis or refraining from doing so (see the next section for details). As a result, ACT's proponents tend to dismiss these two prominent studies as essentially irrelevant, if not fundamentally invalid (e.g. Schultz, 2012; Slantchev, 2012).

My contribution will address these two shortcomings. To do so, I synthesize the case study evidence scattered throughout the literature by performing an analysis that is similar to the meta-analysis of quantitative experimental studies in medicine and psychology

2. For a general discussion of why case studies with process tracing are valuable for testing the empirical validity of formal models, see Goertz (2017: chap. 6) and Lorentzen et al. (2017). Most formal theorists, in political sciences at least, seem to care about the empirical validity of their models. Also, while some of ACT's proponents reject the notion that a model should be subject to empirical testing (e.g., Slantchev, 2017), most ACT proponents and opponents think otherwise, including Fearon (1994b, 2013) himself.

(for a review, see Gurevitch et al., 2018),[3] with an explicit Bayesian logic. Such an analysis, which I call "quasi-meta-analysis" of case studies (QMACS), is a more rigorous method for assessing the overall impact on the validity of a theory by synthesizing the findings from existing case studies. To paraphrase Jeffreys (2007), I ask "Bayesians [to] Learn from Old Data" more rigorously.

My contribution is therefore both empirical and methodological. Empirically, I advance the empirical testing of ACT with case studies using a new method. Methodologically, I introduce a method for synthesizing evidence from case studies in a more formal and rigorous manner. Such a method should be widely applicable to other theories with many case studies.

Due to space limitations, a more detailed and technical discussion of the method of QMACS will be provided elsewhere (Tang, n.d.b). Here, suffice to say that QMACS bears some similarities to, but critically differs from and thus potentially complements, the recently codified method of "large-N qualitative analysis (LNQA)" (Goertz, 2017: chap. 7; Goertz and Haggard, 2023). Briefly, QMACS pools existing case studies regarding the same theory and then analyzes the (lack of) support for the theory by tracking how evidence from all the existing cases supports or weakens the theory with an explicit Bayesian logic.

By comparison, LNQA tests a theory by (re)codifying and (re)examining all the relevant cases for a theory. Hence, QMACS mostly brings together existing empirical evidence, whereas LNQA is mostly a new empirical undertaking. Moreover, QMACS has an explicit Bayesian logic that disciplines the analysis, whereas LNQA can but has yet to explicitly integrate the Bayesian logic (cf. Goertz and Haggard, 2023: esp. 1228–1229).[4]

The QMACS shares some key components with Fairfield and Charman (2022) and Humphreys and Jacobs (2023). Humphreys and Jacobs (2023: 293–298) briefly discuss meta-analysis of quantitative studies with a Bayesian logic via a "hierarchical model." Meanwhile, Fairfield and Charman (2022, chap. 5; see also Fairfield and Charman, 2017, 2019) discuss Bayesian analysis with multiple cases, but only cases within a single study. Both of them employ Van Evera's four types of tests and evidence as a key component. Because QMACS synthesizes evidence from a set of cases by different studies aimed at a single theory with an explicit Bayesian logic, QMACS is a dedicated and more versatile tool for synthesizing case studies for a single theory.

Several caveats are in order before I proceed.

First, I do not deny the enormous impact of Fearon's framework on how to link domestic politics with international politics, especially in crisis bargaining (e.g. Schultz,

3. For a standard textbook of meta-analysis, see Borenstein et al. (2021). It should be noted that meta-analysis of experimental studies has its own critics (e.g., Freedman, 2010, 33-37; Corker, 2022). My position is that when done properly and based on well-designed individual studies, meta-analysis in medicine and psychology does provide us with a valuable tool for synthesizing studies scattered in a field. Here, I use quasi-meta-analysis in a metaphorical sense: the purpose is to reexamine and synthesize different case studies with the same set of requirements for evidence and causal inference. Global sensitivity analysis with observational data is a bit similar to meta-analysis, but the former is based on statistical assumptions that are much less stringent.
4. Both Dion (1998) and McKeown (1999) have showed that case studies can be combined with the Bayesian logic.

2012; Slantchev, 2012). Quite the opposite, this contribution testifies to Fearon's enormous impact. Moreover, I readily admit that audience costs are a valid factor or mechanism that links domestic politics with international politics in crisis bargaining. *What I do dispute is that audience costs are as central or weighty as ACT's proponents have contended* (see also Snyder and Borghard, 2011).

Second, the sizable literature built upon ACT was not Fearon's intention or encouraged by him. In fact, Fearon subsequently moved to other projects and did not even participate in the debate on audience costs. For Fearon, generating audience costs is only one possible ingredient or mechanism for generating credibility and signaling resolve out of many.

Third, this article is not a review of the voluminous ACT literature (for a limited and somewhat dated bibliography online, see Chan et al., 2021), and I apologize for not citing more relevant studies. Relatedly, some of the criticisms I bring up have been leveled against ACT before, though I draw from and extend them whenever possible.

Fourth, my discussion focuses on audience costs in real crises exclusively, as Fearon (1994a, 1994b, 1997) originally intended. As Fearon (1992) noted, some "crises" are merely public stunts or charades that politicians play to both domestic and international audiences. In such situations, ACT may not apply since there is little or no risk of war (pp. 183–186). Also, some authors have extended audience costs to economic treaty negotiations (e.g. Simmons, 2010). Because such situations are not crises, I shall not discuss them either.

Fifth, this article is not a wholesale rejection of formal modeling of crisis bargaining and of international politics in general. Proper formalization can enforce explicit assumptions and produce counterintuitive insights. Rather, the essay seeks to reinforce Jervis's (1989: 184) earlier point that the deductive model still needs some empirical support before being taken seriously (see also Schelling, 1960: 162–164; Mercer, 2012: 402–404; Snyder and Borghard, 2011; for the argument that formal models can do away with empirical validity, see Johnson, 2021; Slantchev, 2017).

Finally, because I focus on empirical studies based on observational data, I do not address attempts to test ACT with online survey experiments. Although experimental studies provide important insight into the microfoundation for ACT and generally support ACT, the external validity of experimental evidence has remained contentious (Barabas and Jerit, 2010), even if cross-cultural differences may be minimal (Bassan-Nygate et al., 2025). While subjects in the experiment can assert that standing firm with one's commitments is a virtue for leaders, subjects as real voters may not punish leaders who escalated and then backed down in the real world. Moreover, subjects in these online experiments are primed to pay attention to audience costs (Snyder and Borghard, 2011, 441; Gartzke and Lupu, 2012; Kurizaki and Whang, 2015: 957–958). As a result, studies based on experiments might have overestimated the size of audience costs. Moreover, even within these experiments, evidence for ACT is inconclusive at best (e.g. Tomz, 2007 vs Levendusky and Horowitz, 2012; Trager and Vavreck, 2011). Furthermore, both experimental studies and case studies have shown that politicians can use various "spinning" tactics for minimizing or even avoiding audience costs altogether (e.g. Hauenstein, 2020; Lin-Greenberg, 2019; Wong, 2024). Indeed, experimental studies have shown that audience costs are highly contingent and thus potentially quite marginal (e.g. Croco et al., 2021; Levy et al., 2015; Nomikos and Sambanis, 2019). Altogether, experimental studies do not conclusively support ACT, to say the least.

The rest of the article proceeds as follows. The next section performs the quasi-meta-analysis of existing case studies of ACT. The next next section identifies three signs that ACT is an unproductive enterprise. A brief conclusion follows.

Case studies of ACT: a quasi-meta-analysis

From Fearon's original formulation, four central hypotheses can be derived. (1) Leaders tend to issue public threats because doing so increases the credibility of one's threat by tying one's hands. (2) All else being equal, in a crisis bargaining situation, the side with (potentially) higher audience costs is more likely to prevail than the other side. And because democracy tends to generate higher audience costs (by escalating publicly), democracy should prevail more frequently than autocracy, all else being equal.[5] (3a) The public in a democracy should punish a leader who publicly escalated a crisis and then backed down, regardless of the substance of the crisis and policy. (3b) The public in a democracy should punish a leader who escalated and then backed down only when the public is hawkish. (This is a weaker version of 3a.) (4) A leader may prefer fighting over backing down when the anticipated audience costs are very large.

Unfortunately, these hypotheses have not been formulated rigorously enough, partly because they have not been classified according to the four types of tests and evidence coined by Van Evera (1997). In contrast, the discussion that follows focuses on reformulating these hypotheses and classifying them according to the four types coined by Van Evera (1997).

Two earlier qualitative studies with process tracing have challenged ACT vigorously (Snyder and Borghard, 2011; Trachtenberg, 2012a). Both studies found the evidence for ACT to be underwhelming. Snyder and Borghard (2011) summarized the gist of their argument succinctly: "We found domestic audience costs as secondary mechanisms in a few cases where the public already had hawkish preferences before any threats were made" (p. 437).

Partly because even ACT's proponents admit that in-depth case studies test ACT more directly than quantitative exercises (Schultz, 2001a: 32–36, 52–55), the two case-centered critiques by Snyder and Borghard (2011) and Trachtenberg (2012a) resulted in a spirited debate in Security Studies (2012), with Schultz (2012) and Slantchev (2012) on one side versus Trachtenberg (2012b) and Mercer (2012) on the other side and Gartzke and Lupu (2012) and Levy (2012) in between. ACT's proponents, however, have largely dismissed the two studies with "selection effects."

Here, it is essential to differentiate "selection effects" from "selection biases."[6] Briefly, "selection biases" are methodological: when a researcher selects only positive or negative cases for his or her theory, selection biases are committed. Similarly, when a researcher selects data that are most likely to support his or her theory from a universe of valid observations, she/he is committing "selection biases." In other words, "selection biases" operate after data generation.

In contrast, "selection effects" in the crisis bargaining literature operate only in situations in which decision-makers strategically select themselves into different situations,

5. This is Fearon's (1994b) original plausible working hypothesis.
6. Indeed, Danilovic (2002) seems to have used these two terms as essentially equivalent.

such as deterrence, crisis, and alliance. In other words, "selection effects" operate in the data-generating process and are not methodological effects. As such, case studies with process tracing are better equipped for alleviating this problem of "selection effects" than quantitative exercises. Indeed, Schultz (2001a: 32–36, 52–55), a major proponent of ACT, has explicitly stressed that case studies with process tracing are better suited for identifying audience costs with selection effects being present.

By comparison, quantitative exercises with observational datasets cannot handle selection effects effectively. Indeed, selection effects may well render many earlier quantitative tests with the militarized international dispute (MID) dataset and the international crisis behavior (ICB) dataset, inconclusive or even inappropriate (see the next section).

Altogether, I perform four tasks in this section. First, I summarize and classify almost all the cases examined in the literature with ACT in mind. This exercise is thus equivalent to a meta-analysis in quantitative studies (see Table 2). By doing so, I provide a more integrative overview of the case-based studies for ACT. More importantly, such a quasi-meta-analysis provides an arguably more rigorous test for ACT's empirical validity. Second, and more critically, I explicate what in-depth case studies with process tracing can uncover to support or weaken ACT. Earlier opponents and proponents of ACT have not explicitly categorized the four types of evidence according to their leverage of differentiating different hypotheses, as Van Evera (1997), Collier (2011), Bennett (2015), and others have articulated. As a result, opponents and proponents of ACT tend to produce less-rigorous conclusions, even though they have addressed key issues regarding case selection, selection effects, and causal inference (e.g. Schultz, 2001a; Slantchev, 2012; cf. Snyder and Borghard, 2011; Trachtenberg, 2012a, 2012b; see also Gartzke and Lupu, 2012; Levy, 2012). Third, I perform the actual QMACS of the cases. Finally, I focus on the Fashoda crisis between France and Britain in 1898, a bread-and-butter case for audience costs (Fearon, 1994b; Schultz, 1998, 2001a). I summarize how ACT's proponents have interpreted the crisis and then show how their interpretations of the crisis unravel in the light of logic with process tracing (see also Trachtenberg, 2012a, 2012b).

Throughout the discussion, I take ACT's proponents' emphasis on selection effects seriously (e.g. Fearon, 2002; Schultz, 2001a, 2012; Slantchev, 2012) and explicitly address the role of selection effects for uncovering evidence for or against ACT (e.g. going public with threats vs going private, see Kurizaki, 2007). By doing so, I shall show that while ACT's proponents have been right to call selection effects to the attention of ACT's critics (e.g. Snyder and Borghard, 2011; Trachtenberg, 2012a), selection effects have not been that prevalent in preventing empirical testing of ACT with rigorous process tracing *because selection effects operate only in the phase of decision to escalate a crisis publicly or not*. As a result, selection effects only apply to "straw-in-the-wind" and "hoop" tests, but not to "smoking gun" and "doubly decisive" tests (see Table 1). Hence, selection effects cannot shield ACT from serious empirical challenges.

I focus on democracies because, according to Fearon (1994b) and his critics, democracies are where audience costs should most likely operate. Hence, if audience costs do not operate or do not matter much even in democracies, then ACT is seriously weakened (cf. Schultz, 2012; Slantchev, 2012). In contrast, there are many possible explanations for why audience costs are not found in autocracies. As a result, causal inference will be more uncertain, if not difficult, with autocracies.

Table 1. Evidence and hypotheses for process tracing audience costs in crises.[a]

Types of evidence	Hypotheses	Effects for supporting ACT versus weakening ACT	Are selection effects involved?	Historical cases as possible examples
Straw-in-the-wind (SW)	SWH-1: a democracy escalates a crisis by issuing public threats voluntarily, especially when facing a nondemocracy. SWH-2: an autocracy backs down when facing a public threat issued by a democracy. SWH-3: a leader, whether in an autocracy or a democracy, is punished after backing down when the other side issues a public threat. SWH-4: war tends to break out more frequently between two states after public threats from one or both sides.	Ambiguous and weak: passing weakly supports ACT while failing weakly weakens ACT.	Yes, but only for SWH-1: some leaders in a democracy may prefer secret diplomacy during crises (e.g. Nixon and Kissinger).	In the Suez crisis, Britain and France escalated into war but backed down after external pressure from the United States and the Soviet Union. After the defeat, Eden resigned, but Mollet stayed in power. Prussia's foreign minister, Alfred von Kiderlen-Waechter (1852–1912), was severely criticized after Prussia's humiliation in the second Morocco (Agadir) crisis.
Hoop (HP)	HPH-1: leaders, especially those in a democracy, tend to make their threats public, especially when facing nondemocracies. HPH-2: leaders, especially those in a democracy, tend to make their public threats as ultimatums or quasi-ultimatums rather than vague threats, especially when facing nondemocracies.	Passing weakly supports ACT while failing strongly weakens ACT.	Yes. HPH-1 is a populational version of SWH-1. The more frequent democracies make threats public, the stronger the support for ACT. Yes. HPH-2 captures the severity of public threats issued by democracies.	The Cuban Missile Crisis: John F Kennedy made a public and quasi-ultimatum to Khrushchev.

(continued)

Table 1. (continued)

Types of evidence	Hypotheses	Effects for supporting ACT versus weakening ACT	Are selection effects involved?	Historical cases as possible examples
Smoking-gun (SG)	SGH-1 (for ACT): leaders in a democracy are punished after backing down from their public threats against another state (democracy or not), *regardless of whether the public initially approves or disapproves leaders' hawkish policies.*[b] SGH-2 (against ACT): leaders in a democracy are not punished after backing down from their public threats against another state (democracy or not) *when the public initially disapproves of leaders' hawkish policies.*	SGH-1: passing strongly supports ACT while failing weakly weakens ACT. SGH-2: passing strongly weakens ACT while failing weakly supports ACT.	No.	The Fashoda crisis: France backed down, and hence its leaders might have been (severely) punished. (See the detailed discussion of the Fashoda crisis in the main text.)
Doubly decisive (DS)	DSH-1 (for ACT): leaders in a democracy are punished after backing down from their public threats against another state (democracy or not), *even though the public has initially disapproved of leaders' hawkish policies.* DSH-2a (against ACT): leaders in a democracy are not punished after backing down *when the public initially disapproves of leaders' hawkish policies.* DSH-2b (against ACT): leaders in a democracy are punished for escalating despite winning the conflict *because the public initially disapproves of leaders' hawkish policies.*	Passing DSH-1 strongly supports ACT while failing severely weakens ACT. Passing DSH-2a or DSH-2b strongly weakens ACT, although cases supporting DSH-2a may be rare because the public may punish leaders out of being humiliated. Cases supporting DSH-2b will be even rarer because "the victors get all the glory."	No.	DSH-1: the Syrian Crisis in 2012: Obama issued a threat against Syria's possible use of chemical weapons (i.e. "the red line") but did not enforce it in 2013. The US public was against military intervention. Examples that support DSH-2b will be extremely rare because "the victors get all the glory." The recent US/Israel war against Iran may provide one such example, esp. for DSH-1 and DSH-2a.

[a]I thank several reviewers for pressing me to be more precise and explicit with these possible hypotheses.
[b]Snyder and Borghard (2011) classify this kind of evidence as smoking-gun evidence (p. 442). They might have used the term in a colloquial way. Moreover, they mention only this type of evidence or test.

Four types of evidence from process tracing and their Bayesian logic

According to Van Evera's (1997) colorful labels, there are four types of evidence that can be derived from case studies with process tracing: (1) "straw-in-the-wind (SW)," (2) "hoop test (HP)," (3) "smoking-gun (SG)," and (4) "doubly decisive (DS)" (see also Bennett, 2015; Collier, 2011). Because several different hypotheses can be derived from a theory and different hypotheses can be subjected to different tests, the validity of a theory is a function of the overall support that its hypotheses receive in different tests, subject to Bayesian updating (e.g. Beach and Pedersen, 2016, 2019; Bennett, 2015; Collier, 2011; Fairfield and Charman, 2022; Humphreys and Jacobs, 2023; Rohlfing, 2014).

The logic of Bayesian updating is straightforward and intuitive. For simplicity, I shall only address the situation in which a hypothesis is either true or false. Let h denote hypothesis, e evidence, and p probabilities. The $p(h)$ is the prior probability that one's empirical hypothesis is true, and $p(\sim h)$ is the prior probability that one's hypothesis is false. The $p(e \mid h)$ is the probability that one finds e if one's h is correct, and $p(e \mid \sim h)$ is the probability that one finds e if one's ~h is correct (or h is incorrect). The posterior Bayesian probability after finding a piece of evidence e, $p(h \mid \mathrm{e})$, is determined by the Bayesian theorem:

$$p\left(h|\mathrm{e}\right) = \frac{p(h)}{p\left(h\right) + \frac{p(e \mid \sim h)}{p(e \mid h)}\, p(\sim h)}$$

Note that $p\left(h\right) + p\left(\sim h\right) = 1$ and $p\left(e|h\right) + p\left(e| \sim h\right) = 1$. As a result, evidence enters into the Bayesian probability equation by changing the value of $\frac{p(e \mid \sim h)}{p(e \mid h)}$. With the Bayesian probability equation, we can now link the four types of evidence with posterior Bayesian probability (Bennett, 2015; Fairfield and Charman, 2019, 2022), via their "differentiating power."[7]

Straw-in-the-wind evidence is the weakest among the four because it has little "differentiating power": finding such evidence provides little support for a hypothesis, while failing to find such evidence weakens the hypothesis only slightly. In other words, straw-in-the-wind evidence, either positive or negative, does not significantly change the value of $\frac{p(e \mid \sim h)}{p(e \mid h)}$.

Hoop evidence has more differentiating power than straw-in-the-wind evidence. Passing a hoop test provides weak support for a hypothesis, while failing it eliminates or severely weakens a hypothesis. In other words, passing a hoop test reduces the value of $\frac{p(e \mid \sim h)}{p(e \mid h)}$ somewhat, thus slightly increasing the value of $p\left(h|\mathrm{e}\right)$. In contrast, failing a hoop test greatly increases the value of $\frac{p(e \mid \sim h)}{p(e \mid h)}$, thus significantly decreasing the value of $p\left(h|\mathrm{e}\right)$.

7. A more detailed and technical discussion of "differentiating power" will be provided elsewhere (Tang, n.d.b).

A smoking-gun test is the converse of a hoop test: passing a smoking-gun test provides strong support for a hypothesis, while failing a smoking-gun test weakens the hypothesis slightly. In other words, finding positive smoking-gun evidence greatly decreases the value of $\frac{p(e|\sim h)}{p(e|h)}$, thus significantly increasing the value of $p(h|e)$. Yet, failing to find smoking-gun evidence does not significantly change the value of $\frac{p(e|\sim h)}{p(e|h)}$ and hence the value of $p(h|e)$.

Finally, the doubly-decisive test provides the most differentiating power: passing such a test confirms a hypothesis while eliminating other competing hypotheses, whereas failing such a test eliminates or severely weakens a hypothesis. Put it differently, doubly-decisive evidence, whether positive or negative, significantly changes the value of $\frac{p(e|\sim h)}{p(e|h)}$, thus greatly impacting the value of $p(h|e)$. Positive doubly-decisive evidence significantly increases the value of $p(h|e)$, whereas negative doubly-decisive evidence significantly decreases the value of $p(h|e)$.

Strong process tracing-based evidence against ACT

Because case studies with process tracing test ACT's central hypotheses directly (Schultz, 2001a), if case studies consistently uncover evidence supporting ACT, ACT will be significantly strengthened. In contrast, if case studies consistently fail to find supporting evidence, then ACT will be weakened, at least somewhat. Furthermore, if case studies uncover powerful evidence against ACT, then ACT will be seriously weakened.

For ACT, evidence for the following hypotheses is straw-in-the-wind evidence: (SWH-1), a democracy escalates in a crisis by issuing a public threat voluntarily, especially when facing a nondemocracy; (SWH-2), an autocracy backs down when facing an explicit threat issued by a democracy; (SWH-3), a leader, whether in an autocracy or a democracy, is punished after backing down; and (SWH-4), war breaks out between two states after public threats from one or both sides (regardless of whether the two states are democracies or not). Straw-in-the-wind evidence has very low differentiating power, most critically because the evidence might have been driven by other causes (Snyder and Borghard, 2011: 411). I therefore skip such evidence below.

For ACT, evidence for the following hypotheses is hoop evidence: (HPH-1) leaders, especially those in a democracy, tend to make their threats public, especially when facing nondemocracies; and (HPH-2) leaders, especially those in a democracy, tend to make their public threats as ultimatums or quasi-ultimatums rather than vague threats, especially when facing nondemocracies. Note that HPH-1 is a populational version of SWH-1: the more frequent democracies make threats public, the stronger the support for ACT.

Again, hoop evidence only supports weakly but does not confirm ACT because leaders (democratic or not) may make public threats for considerations other than enhancing credibility by incurring potential postcrisis audience costs. In fact, leaders may make public threats because they wish to rally public support or precisely because they wish to avoid potential postcrisis audience costs: public threats may be based on wishful

thinking (Lebow, 2020 [1981b], esp. chap. 7). Lack of such evidence, however, seriously weakens but does not disprove ACT (Collier, 2011).

Evidence for the following hypotheses is smoking-gun evidence. The SGH-1 (for ACT): leaders in a democracy are punished after backing down from their public threats against another state (democracy or not), *regardless of whether the public initially approves or disapproves of leaders' hawkish policies*. Against ACT, evidence for the following hypotheses is smoking-gun evidence. SGH-2 (against ACT): leaders in a democracy are *generally* not punished after backing down from their public threats against another state (democracy or not) *when the public initially disapproves of leaders' hawkish policies*.

For ACT, evidence for the following hypothesis is doubly-decisive evidence. The DSH-1 (for ACT): leaders in a democracy are more likely to be punished after backing down from their public threats against another state (democracy or not), *even though the public has initially disapproved of leaders' hawkish policies*. Such evidence provides strong support for ACT, while it seriously weakens a key critique of ACT, that is, the public cares more about the substance of the policy than leaders' faces or their country's honor.[8]

Against ACT, evidence for the following hypotheses is doubly-decisive evidence. The DSH-2a (against ACT): leaders in a democracy are more likely to be rewarded after backing down, *when the public initially disapproves of leaders' hawkish policies*. The DSH-2b (against ACT): leaders in a democracy are more likely to be punished for escalating despite winning the conflict, *because the public initially disapproves of leaders' hawkish policies*.

Because doubly-decisive evidence has the most differentiating power, finding supporting evidence for ACT strongly strengthens ACT, while seriously weakening a key critique of ACT (i.e. that the public cares more about the substance of the policy than leaders' faces or their country's honor). By the same logic, finding confirming evidence against ACT (or disconfirming evidence for ACT) seriously weakens ACT while strongly strengthening a key critique of ACT. Of course, doubly decisive evidence for both DSH-2a and DSH-2b may be rare because the public may punish leaders for being humiliated (for DSH-2a) or because of the "rally-around-the-flag" (for DSH-2b).

Intuitively, excluding the straw-in-the-wind evidence, if the remaining three sets of evidence mostly go against ACT, then our confidence in ACT will be seriously weakened, according to the intuitive logic of Bayesian updating. In contrast, if the three sets of evidence mostly support ACT, then our confidence in ACT will be greatly strengthened.

Table 1 summarizes these four types of evidence and their corresponding hypotheses.

A key point is worth being repeated: selection effects only apply to "straw-in-the-wind" and "hoop" tests but not to "smoking gun" and "doubly decisive" tests (see Table 2 below). This is so because leaders only get to choose their actions in escalating and backing down, while the public decides whether to punish or reward leaders after the result of a crisis becomes fixed (i.e. backing down, fighting, winning, losing, or

8. Snyder and Borghard (2011: 442) classify this set of evidence as smoking-gun evidence. They might have used the term in a colloquial way. Moreover, they only mention this set of evidence or tests.

Table 2. Summary of evidence from case studies by various authors.

Name of crisis	Hoop evidence: did democracy issue strong public threats and thus generate significant audience costs voluntarily?	Smoking-gun evidence for ACT: did the public punish democratic leaders after their backing down, regardless of the public's initial position regarding the dispute?	Doubly decisive evidence-1 *for* ACT: leaders in (mature) democracy are punished after backing down, *even though the public initially disapproved of leaders' hawkish policies.*	Doubly decisive evidence-2 *against* ACT: leaders in (mature) democracy are not punished after backing down or are punished after escalating.
The Suez crisis (1956)	Strong Yes: with a public ultimatum	Ambiguous: both the British public and the French public supported the intervention. Eden resigned shortly after, but the Conservative government stayed. Mollet stayed in power.	N. A.: both the British public and the French public initially supported the intervention.	Ambiguous: Britain and France escalated into war but backed down after external pressure. After the defeat, Eden resigned, but Mollet stayed in power.
The Iran hostage crisis (1979–1980)	Weak Yes: public but not so explicit threat	N. A.: the public supported Carter's policy. They punished him for being "incompetent."	Weakly against: the public, punished Carter, but mostly for his "dovishness" and "incompetence," not his "inconsistency."	N. A. or ambiguous: the public later wanted more hawkish policies.
The Sino-India crisis and border war (1959–1960)	No: no public and explicit threat	N.A.: India did not back down.	N.A.: India did not back down.	N. A.: India did not back down.
The Cuban Missile Crisis (1962)	Yes, with a public quasi-ultimatum	N. A.: the United States did not back down.	N. A.: the United States did not back down.	N. A.: the United States did not back down.
The Eastern crisis (1877–1878)	Yes: public and explicit threat without ultimatum	N. A.: Britain did not back down.	N. A.: Britain did not back down.	N. A.: Britain did not back down.
The Fashoda crisis (1898)	No: Salisbury made the dispute and threat public quite reluctantly.	No: French leaders were not punished despite backing down.	N. A.: the French public supported French hawkish policies.	Yes: French leaders were not punished despite backing down.
The first Moroccan crisis (1905–1906)	Strong No: France made no public threats and actually signaled weakness.	N. A.: France did not back down.	N. A.: France did not back down.	N. A.: France did not back down.
The second Moroccan/ Agadir crisis (1911)	Weak Yes: public but not so explicit threat	N. A.: Britain did not back down.	N. A.: Britain did not back down.	N. A.: Britain did not back down.
The Turkish crisis (1946)	No: no public and explicit threat	N. A.: the United States did not back down.	N. A.: the United States did not back down.	N. A.: the United States did not back down.
The Iranian crisis (1946)	Weak Yes: public but not so explicit threat	N. A.: the United States did not back down.	N. A.: the United States did not back down.	N. A.: the United States did not back down.

(continued)

Table 2. (continued)

Name of crisis	Hoop evidence: did democracy issue strong public threats and thus generate significant audience costs voluntarily?	Smoking-gun evidence for ACT: did the public punish democratic leaders after their backing down, regardless of the public's initial position regarding the dispute?	Doubly decisive evidence-1 *for* ACT: leaders in (mature) democracy are punished after backing down, *even though the public initially disapproved of leaders' hawkish policies.*	Doubly decisive evidence-2 *against* ACT: leaders in (mature) democracy are not punished after backing down or are punished after escalating.
The first Berlin crisis (1948–1949)	No: no public explicit threat	N. A.: the United States did not back down.	N. A.: the United States did not back down.	N. A.: the United States did not back down.
The second Berlin crisis (1958–1962)	No: no public explicit threat	N. A.: the United States did not back down.	N. A.: the United States did not back down.	N. A.: the United States did not back down.
The remilitarization of the Rhineland (1936)	Weak Yes: public threat, but no ultimatum	No: the public did not punish either British or French leaders.	N. A.: the British and French public supported dovish policies.	Strong Yes
The Czechoslovakia/ Munich crisis (1938)	Weak Yes: public threat, but no ultimatum	No: the public did not punish either British or French leaders.	N. A.: the British and French public supported dovish policies.	Strong Yes
The Abadan crisis (1951)	Weak Yes: public and explicit threat, though not an ultimatum.	No: the public did not punish British leaders after backing down.	N. A.: the British public supported dovish policies.	Strong Yes. The Labor Party was rewarded.
The Vietnam crisis (1954)	Yes, with public threats	N. A.: the United States did not back down.	N. A.: the United States did not back down.	N. A.: the United States did not back down.
The Vietnam decision (Johnson in 1965)	Yes, with public threat	N. A.: the United States did not back down.	N. A.: the United States did not back down.	Strong Yes: Johnson was punished for escalating.
The withdrawal from Vietnam (Nixon & Kissinger, 1972–1973)	Yes, with public threat	Against ACT: the public did not punish Nixon and Kissinger. In fact, they rewarded them.	N. A.: the United States public supported de-escalation.	Strong Yes: the public rewarded Nixon and Kissinger.
The Schleswig-Holstein crisis (1863–1964)	No: Palmerston issued a public warning against Prussia and Austria despite knowing Britain's relative weakness.	No: the public did not punish British leaders after backing down.	Ambiguous: the public did not punish British leaders after backing down, but the parliament censored Palmerston.	Yes: the public and media were supportive of Palmerston after his being censored.

(continued)

Table 2. (continued)

Name of crisis	Hoop evidence: did democracy issue strong public threats and thus generate significant audience costs voluntarily?	Smoking-gun evidence for ACT: did the public punish democratic leaders after their backing down, regardless of the public's initial position regarding the dispute?	Doubly decisive evidence-1 *for* ACT: leaders in (mature) democracy are punished after backing down, *even though the public initially disapproved of leaders' hawkish policies.*	Doubly decisive evidence-2 *against* ACT: leaders in (mature) democracy are not punished after backing down or are punished after escalating.
Summary of the 19 cases	Uncertain: ACT passes hoop tests convincingly in only 6 cases and weakly in 6 cases but fails in 7 cases out of a total of 19 cases.	ACT failed "smoking-gun" tests in all seven relevant cases, with one strong against ACT. The other 12 cases are irrelevant because democracies did not back down.	One weakly against ACT, one ambiguous. The other 17 cases are irrelevant because democracies did not back down.	Of the eight relevant cases, seven relevant cases support ACT's critics, with five very strongly. The other 11 cases are irrelevant.

N. A.: not applicable. This table merely summarizes the case studies by different authors with ACT in mind, without going through firsthand (archive) or even secondhand (i.e. historical accounts) materials. The first four crises are from Snyder and Borghard (2011). The next eight are from Trachtenberg (2012a). The three cases about Vietnam were from Trachtenberg (2014a, 2014b). See also McManus (2017, chap. 8). The Abadan crisis (1951) is from Schultz (2001a: 42–45), whereas the Boer War crisis (1899) is from Schultz (2001b: 200–206). Schultz (2001b: 175–195, 206–223) also examined the Fashoda crisis, the Rhineland crisis, and the Suez crisis. The Schleswig-Holstein crisis (1863–1864) is from Schlesinger and Levy (2021). Ramsay (2004) also examined the Suez crisis. The "war-in-sight" crisis (1871) between Prussia and France/Britain (Trachtenberg, 2012a) and the Rhodesian independence crisis (1965, Schultz, 2001b: 223–228) are irrelevant cases because they provide very limited leverage for testing ACT. All authors mentioned above have examined the Cuban Missile Crisis, as did Brown and Marcum (2011).

Fearon (1992: 144–151) briefly mentioned the Fashoda crisis, the India–China crisis, the Second Berlin crisis, the Cuban Missile Crisis, and the Agadir crisis as anecdotal evidence for supporting his "plausible working hypothesis." Fearon (1992) also faulted the regime as the key cause behind why Mao's warning against Truman had failed to deter the United States from advancing to the Yalu River and Saddam's intransigency had failed to deter Bush from launching "Desert Storm" (pp. 147–151).

Slantchev (2011) also examined the India–China crisis (chap. 5) and the Korean War crisis between China and the United States (chap. 6).

Snyder and Diesing (1977), Lebow (2020 [1981b]), and George and Simons (1994) have examined some of these cases (e.g. the Fashoda crisis and the Sino-India crisis) and many other crises before the coming of ACT. See also Freedman (1998).

stalemate). As a result, selection effects cannot shield ACT from serious empirical challenges.

A related point is that a case may only contain one, two, or three types of tests rather than all four types. This is partly due to the presence of selection effects but also due to other factors that impact whether audience costs will materialize and how high the costs are if materialized. For instance, when the public opposes a leader's hawkish position, the former may not punish the leader if the latter eventually backs down. Indeed, the public may reward the leader for being prudent.

We now move to the actual meta-analysis of the relevant case studies by coding the cases and evidence according to the four types of evidence and hypotheses. Table 2 summarizes the findings from Schultz (2001a, 2001b), Snyder and Borghard (2011), Trachtenberg (2012a, 2014a, 2014b), and others by explicitly categorizing the evidence according to the three types of evidence noted above. In order to directly test ACT's central theses, each case involves at least one democracy.

From Table 2, ACT passes hoop tests (strong plus weak) in 12 out of the 19 cases and fails in 7 cases. ACT thus gains only weak support from hoop tests. ACT, however, has clearly failed in all but one of the smoking-gun tests for ACT (the 1956 Suez crisis) in the 19 cases, even though there is only one smoking-gun case against ACT (the withdrawal from Vietnam by Nixon and Kissinger in 1972–1973).

More importantly, however, in the doubly-decisive tests against ACT, of the eight relevant cases, only one is ambiguous (i.e. the Suez crisis), while seven clearly go against ACT: the public did not punish leaders who had backed down when the public was against escalation, even though audience costs should be quite high in some cases (e.g. the Abadan crisis). By comparison, there is no doubly-decisive case for ACT. Altogether, evidence from these case studies does not support ACT but has seriously weakened it.

For illustrative purposes, we deploy the Bayesian theorem to perform a simple overall assessment of ACT based on our QMACS. Because failing to find smoking-gun evidence does not significantly change the value of $\frac{p(e|\sim h)}{p(e|h)}$ and hence the value of $p(h|\mathrm{e})$. We disregard the lack of confirming evidence for ACT from smoking-gun tests. We are now left to work with evidence from hoop tests and doubly-decisive tests.

To give ACT the benefit of the doubt, let us assume the prior probability that ACT is correct (i.e. $p(h)$) to be 0.6. Regarding hoop tests, let us be generous with ACT and assume that ACT passes 12 tests out of 19 cases but fails in only 7 out of 19 cases. This means the probability of $p(e|h)$ is 12/19, whereas $p(e|\sim h)$ is 7/19. It can be easily shown that, based on hoop tests, the posterior probability that ACT is correct is now merely 0.72 (i.e. [$p(h|\mathrm{e}) = \frac{0.6}{0.6 + \frac{7}{12}(0.4)} = 0.72$]). Consistent with the Bayesian logic with hoop tests, hoop tests seem to provide more support for ACT, starting with a baseline of $p(h) = 0.6$. Hoop tests, however, are a weak test: passing them does not necessarily entail that the hypothesis is correct.

We next move to doubly-decisive tests. Again, let us be generous with ACT by disregarding the fact that ACT receives little or no doubly-decisive support. Rather, we focus on the evidence from doubly-decisive tests that are against ACT. Out of eight relevant cases, seven go against ACT while one is ambiguous. It can be easily shown that, based

on this set of evidence, the posterior probability that ACT is correct is now merely 0.176 (i.e. $[\, p(h|e) = \frac{0.6}{0.6 + \frac{7}{1}(0.4)} = 0.176\,]$). Thus, the prospect of ACT looks rather dim with the doubly-decisive test against it alone.

Finally, we combine the evidence from hoop tests and doubly decisive tests against ACT. If we let hoop tests go first and doubly decisive tests second, the posterior probability that ACT is correct is now merely 0.268 (i.e. $[\, p(h|e) = \frac{0.72}{0.72 + \frac{7}{1}(0.28)} = 0.268]$). Alternatively, we can let doubly decisive tests go first and hoop tests second; the posterior probability that ACT is correct is again merely 0.268 (i.e. $[\, p(h|e) = \frac{0.176}{0.176 + \frac{7}{12}(0.824)} = 0.268\,]$).[9] Things look quite bleak for ACT.

Thus, after subjecting the cases to process tracing with an explicit Bayesian logic and then synthesizing the evidence with the more rigorous procedure of QMACS with an explicit Bayesian logic, a clear picture emerges: the validity of ACT is in serious doubt. Moreover, as becomes clearer in the Fashoda crisis below, neither "selection effect" nor "strategic interaction" can explain why audience costs cannot be observed in action (cf. Schultz, 2012; Slantchev, 2012).

The Fashoda crisis reexamined

The Fashoda crisis has been extensively examined, and many ACT proponents have claimed that this case supports ACT's core logic. Most prominently, Schultz's (2001a) major reformulation of ACT centered on the Fashoda crisis as the key supporting case (Schultz, 2001a: 175–195).

To some extent, the Fashoda crisis has been a bread-and-butter crisis for ACT. A major reason behind this is that the two states within the crisis were both democracies. As such, audience costs should operate in both countries. Thus, the Fashoda crisis is a most likely case for ACT. As such, one would expect some "smoking gun" or even "doubly decisive" evidence for ACT from it. Table 3 details the tests and the hypotheses of the crisis and the results of these tests.[10]

So, does the Fashoda crisis really support ACT? The answer is no when examined with our more rigorous procedure disciplined by Bayesian logic.

9. This procedure is similar to what Fairfield and Charman (2017, 2019, 2022) have described, although they focus on causal inference within a single, case but with different pieces of evidence. By comparison, my exercise is about testing a single theory with multiple case studies, disciplined by the Bayesian logic. As Fairfield and Charman have noted, when taking together, the sequence of different types of evidence makes no difference.
10. I thank the two reviewers of the journal for demanding me to make the procedure of doing QMACS more concrete and thus more accessible before a more detailed and technical exposition of QMACS is out.

Table 3. The Fashoda crisis: tests, hypotheses, and evidence.

Tests	Hypotheses	Evidence
Hoop	HPH-1: leaders, especially those in a democracy, tend to make their threats public, especially when facing nondemocracies.	No. Salisbury made the crisis public quite reluctantly. In fact, he only published the diplomatic communications because the domestic pressure made him do so.
	HPH-2: leaders, especially those in a democracy, tend to make their public threats as ultimatums or quasi-ultimatums rather than vague threats, especially when facing nondemocracies.	No. Britain did not issue an ultimatum: instead, it put its navy on high alert to force France to back down.
Smoking-gun	SGH-1 for ACT: leaders in a democracy are punished after backing down from their public threats against another state (democracy or not), *regardless of whether the public initially approves or disapproves of leaders' hawkish policies.*	No. French leaders (e.g. Théophile Delcassé) were not punished despite backing down (and being humiliated).
	SGH-2 against ACT: not applicable.	
Doubly decisive *for* ACT	DSH-1: leaders in a democracy are punished after backing down, *even though the public has initially disapproved of leaders' hawkish policies.*	Not applicable: because the French public supported France's initial hawkish policies.
Doubly decisive *against* ACT	DSH-2a: leaders in a democracy are not punished after backing down *when the public initially disapproves of leaders' hawkish policies.*	Not applicable: because the French public supported France's initial hawkish policies.
	DSH-2b: leaders in a democracy are punished for escalating despite winning the conflict, *because the public initially disapproves of leaders' hawkish policies.*	Not applicable, because France backed down.

For more general statements of these hypotheses, see also Table 1.

In July 1898, a small French expeditionary force led by Captain Jean-Baptiste Marchand reached Fashoda in what is now Kodok of South Sudan. In September 1898, a much larger British-Egyptian force led by Lord Herbert Kitchener arrived in Fashoda, and a standoff ensued. On 3 November, France backed down, and the crisis was over. Importantly, during the entire crisis, both Britain and France were (quasi-)democracies.

The realist explanation for the crisis's outcome is straightforward: France backed down because Britain was the stronger side (Snyder and Diesing, 1977: 123–124; Layne, 1994: 28–33). Arguing against the realist argument and for ACT, Schultz (2001a: 40–42; 2001b: 175–195) asserted that realism cannot explain why the Fashoda crisis materialized in the first place and then identified the crisis as a central case supporting ACT because Britain was able to coerce France into backing down by generating

higher audience costs. Unfortunately, in the light of our more rigorously reformulated hypotheses with Bayesian logic, the Fashoda crisis strongly contradicts ACT (see also Trachtenberg, 2012a).

To begin with, France's *ex ante* knowledge of Britain's overall advantage in power and interest in Egypt and Sudan did not deter France from initiating its adventure in the first place. This fact alone contradicts ACT's starting assumption. According to Fearon (1994a, 1997) and Schultz (2001a), before a challenger initiates a dispute, it should have taken the defender's power and interest into consideration. France's *ex ante* estimation of Britain's power and interest, thus, should have a self-selection effect that should either deter France from initiating the crisis, or, if France did initiate the dispute, then it would be quite unlikely to back down. In other words, France would have been a very resolute challenger precisely because it had challenged Britain. Moreover, because both France and Britain were democracies, France should have anticipated the possible impact of audience costs upon Britain's actions before initiating the challenge. Yet, French leaders apparently were caught off guard and deeply distraught by Salisbury's publication of the diplomatic communications between the two states. This fact again contradicts the logic of ACT.

For realists with at least some awareness of psychological factors, France's action was a typical manifestation of Machiavellian statecraft: it was a "limited probe" that aims to gain something but often results in cutting one's losses when the odds against one become too great (Levy, 2012, 386–387; cf. Schultz, 2001b). Moreover, there might have been some wishful thinking behind France's calculation. France was keenly aware of Britain's interest in Egypt and overwhelming (local or total) power advantage and never intended to fight against Britain. What France aimed to achieve with its adventure was to provoke and then induce Britain to the negotiation table, hoping Britain might be willing to offer some concessions in Africa to France in secret negotiations.

Thus, when France took the probing action, it was unsure of Britain's resolve. When Britain signaled its resolve by putting the Royal Navy on alert, France yielded. Indeed, Schultz (2001b: 183–186) admitted this, noting that France's probing was reasonable when it was *unsure* of Britain's resolve, which is at least partly underpinned by its interest and power (Slantchev, 2011).

When we examine the Fashoda crisis with the more rigorously formulated four types of tests by Van Evera (1997), it becomes abundantly clear that the crisis contradicts ACT decisively. Table 3 summarizes the tests, hypotheses, and evidence. Again, straw-in-the-wind tests are left out.

First, according to the HPH-1 hypothesis, Britain (and Salisbury) should have made the crisis public eagerly and early on in order to compel France to back down by incurring possible audience costs. Also, according to the HPH-2 hypothesis, Britain (and Salisbury) should have made a public ultimatum or semiultimatum in order to compel France to back down by incurring possible greater audience costs.

Yet, despite Britain's overwhelming power advantage, Salisbury was initially *reluctant* to back France into a corner by letting the crisis become public; he only published the diplomatic communications because the domestic pressure made him do so (Schultz, 2001b: 182–183; Trachtenberg, 2012a: 13–17). Thus, Salisbury was not acting according to but against the logic of ACT. Certainly, when leaders can choose to play with audience costs or not (thus making audience costs conditional), the logic of ACT weakens considerably, as both Snyder and Borghard (2011) and Trachtenberg (2012a) have argued.

Moreover, Salisbury did not issue a public ultimatum or even a semiultimatum to France in order to compel France to back down. Thus, ACT did not even pass the HP tests in a pillar case.

Second, ACT also fails in cases in which one should expect it to easily pass the SG tests. According to SGH-1, Théophile Delcassé should be severely punished after being humiliated by Britain. (SGH-2 against ACT is not applicable because the French public was initially supporting France's hawkish stand.)

Yet, according to Schultz (2001b: 186–195) himself, France's leading politicians (e.g. Delcassé) were not punished after backing down and being humiliated. Contrary to Schultz (2001b), the Fashoda crisis thus did not support but in fact decisively contradicted ACT's logic (Trachtenberg, 2012a, 2012b). Certainly, focusing on the prevailing side's actions as supporting ACT while disregarding the other side's actions that may contradict ACT is a choice not easily justified (e.g. Schultz, 2001a, 2001b). Indeed, when given the chance to rebut Trachtenberg (2012a), Schultz (2012: 372) did not object to the former's critique of his interpretation of the Fashoda crisis. Instead, Schultz (2012) sidestepped and merely insisted that ACT is a highly useful theory.

Because France eventually backed down and the French public initially supported France's hawkish positions, doubly-decisive tests (i.e. DSH-1 for ACT and DSH-2a and 2b against ACT) do not apply to the Fashoda crisis. There is, however, another piece of evidence contradicting ACT. After the crisis broke out, France too tried to manipulate public opinion even though it was playing with an extremely weak hand (Schultz, 2001a: 191–195). Thus, between the two democracies in the crisis, the stronger and more resolute side—that is, Britain—did not believe that generating audience costs was necessary and perhaps even deemed it counterproductive for resolving the crisis. In contrast, the weaker side—France—did try to manipulate public opinion even though it intended to back down eventually. Eventually, the side with more power and resolve prevailed, handily.

In sum, when being examined with a more rigorous method disciplined by the Bayesian logic, the Fashoda crisis, a most likely case for ACT, does not support ACT. Certainly, there is no smoking-gun evidence for ACT. Indeed, there is doubly-decisive evidence against ACT (i.e. French leaders were not punished despite backing down and hence being publicly humiliated). When ACT fails in the most likely case for it, our confidence in ACT's validity takes a serious hit.

Here, it is useful to note that the Abadan crisis between Britain and Iran (Persia), another case examined by Schultz (2001a: 42–44), too strongly contradicts ACT as a doubly-decisive case. According to Schultz and Goemans (2019), because Britain was not resolute enough, it issued only limited demands upon Iran. Eventually, the Labor Party government led by Clement Attlee backed down from the crisis. Very critically, the crisis unfolded only 3 weeks before the British election day. Thus, according to the logic of ACT, (potential) audience costs for Attlee and his cabinet should have been quite high, contrary to Schultz (2001a: 42). After all, Smith (1998) explicitly emphasized that replacing an incompetent or inconsistent leader must be the principal mechanism through which audience costs operate (see also Fearon, 1994b). In fact, the opposition party, including Winston Churchill, explicitly tried to make the crisis a key issue in the election, "accusing Attlee of breaking his word of safeguarding the British presence in Abadan" (Beck, 2006: 526). Churchill's tactics thus explicitly invoked the logic of ACT.

Yet, according to Schultz (2001a: 43–44), rather than being punished, the Labor Party might have gained political support from the crisis, partly due to their adroit labeling of the Conservative Party as warmongers but perhaps more due to the British public's opposition against escalation. The Abadan crisis thus suggests that the public cares more about the substance of policies in crises (such as competence) rather than merely consistency. Moreover, leaders can spin things, thus substantially dampening the bite of audience costs. In fact, according to Schultz (2001a: 43) himself, the Labor government "developed a shrewd political strategy for minimizing the electoral repercussions of backing down" even before the actual backing down!

Summary

Altogether, although Schultz (2012) and Slantchev (2012) simply dismissed evidence against ACT from case studies, while Gartzke and Luputu (2012) questioned the damages done by the case studies' evidence against ACT, case studies have, in fact, done serious damage to ACT when understood properly according to the typology of evidence by Van Evera (1997) and Collier (2011).

Perhaps a counterfactual will drive the point home: what would have been the response from proponents of ACT if case studies found overwhelmingly, or even merely inconclusive, supporting evidence for ACT? Proponents of ACT would have declared victory long ago. Instead, proponents of ACT are still scratching their heads for support, as Mercer (2012, 388–389) quipped!

The QMACS exercise, by making different hypotheses more explicit and classifying them according to the four types of evidence with an explicitly Bayesian logic, also sheds new light on individual cases. As shown above, some cases that have been argued to either support or contradict ACT only weakly actually significantly weaken ACT because they do constitute "smoking gun" or "doubly decisive" tests against ACT (e.g. the Fashoda crisis). Meanwhile, neither the Indian-China crisis nor the Cuban Missile Crisis constitutes a powerful (i.e. "smoking gun" or "doubly decisive") test for ACT. In the 1961–1962 India–China crisis, the democracy (i.e. India) was the challenger, while the autocracy (i.e. China) was the defender. During the whole crisis leading to the border war, Nehru did not issue public threat against China, and much of the audience cost for Nehru had been generated by the Indian public and its parliament, even though his decision to publish the diplomatic communications in September 1961 between the two countries had certainly backed him into a corner from which he could not retreat (Lebow 1981a: 184–192; Maxwell, 1970).[11] Moreover, Nehru was punished not for backing down (he did not) but for losing the war.

The Cuban Missile Crisis, while certainly a momentous event and possibly a turning point of the Cold War, provides only hoop test evidence, thus weak support for ACT, because John F Kennedy (JFK) did issue a (quasi-)ultimatum and then held firm against Khrushchev. Yet, the crisis was resolved not solely due to JFK's (quasi-)ultimatum but more due to JFK's military mobilization, imposing a blockade, and willingness to compromise (Lebow and Stein, 1994). Hence, it is difficult to attribute much of a role for

11. Here, one cannot fail to notice the striking parallel between Salisbury's action and Nehru's action.

audience costs in the crisis bargaining process. Moreover, the fact that Khrushchev took enormous risks with full knowledge of America's overwhelming nuclear and conventional superiority in the Americas seriously contradicts one of ACT's assumptions that a challenger should be self-deterred from initiating a challenge (Lebow, 2020 [1981b]: xv). In fact, Brown and Marcum (2011) concluded that the crisis contradicted ACT because autocracy seems to be able to sanction a leader (i.e. Khrushchev) who had backed down more effectively.

Remarkably, most ACT's proponents have chosen to ignore the Munich crisis, in which two democracies (i.e. Britain and France) backed down calamitously when facing an autocracy (i.e. Nazi Germany). Although leaders of Britain and France might have backed down mostly due to their domestic division, military weakness, the need to buy time, and ideological antipathy against the Soviet Union (Layne, 2008; Ripsman and Levy, 2008; Schweller, 2006), the fact that they did not pay any audience costs suggests that ACT has failed in a landmark case. The 1936 Rhineland crisis is even more damaging for ACT (Trachtenberg, 2012a: 32–37), because both Britain and France made explicit threats against Germany and eventually backed down. Both cases suggest that when the public and most elites in a country do not want war, backing down incurs only minimal domestic audience costs. In other words, the public and elite often care far more about the substance of a policy than about a leader's courage and consistency or even the honor of their state.

ACT as an unproductive program? Three signs

The preceding sections show that supporting evidence for ACT has been rather disappointing. Contrary to Fearon's (1994b) implicit thesis that audience costs constitute a major mechanism in crisis bargaining, audience costs may not matter that much in crisis bargaining.

When facing growing empirical anomalies, a (productive) research program has essentially two ways to respond: (1) by demonstrating that empirical anomalies are not anomalies or that evidence is invalid or (2) by modifying the theory to accommodate challenging evidence without violating the theoretical core. Unfortunately, ACT's proponents have so far failed to do either. As such, ACT has become an unproductive enterprise, as indicated by three key signs.

The first sign is that when responding to their critics, ACT's supporters simply ignore them or assert that criticisms of ACT were wrong, without providing any concrete evidence. For instance, Schultz (2001b: 175–195) asserted that the Fashoda crisis was consistent with ACT. After Trachtenberg (2012a) challenged his interpretation of the crisis, and Schultz was given the opportunity to respond, however, Schultz (2012) did not even mention the Fashoda crisis but merely insisted that "audience costs" is a useful idea that we should not reject.

Alternatively, ACT's supporters simply assert that their critics' evidence is irrelevant because it does not impinge upon ACT's core. Most prominently, Slantchev (2012) argued that evidence that democracies do not prevail more in crises or that leaders in democracies are not punished after backing down is irrelevant for ACT. Such a stand is disingenuous, at least somewhat. Although Fearon (1994a, 1997) did not argue that only a democracy can generate audience costs, he explicitly noted that democracies tend to

generate higher audience costs more easily and thus are more likely to prevail. Thus, the fact that democracies do not prevail more in crises or that leaders in democracies are not punished after backing down does cast serious doubt against the validity of one of ACT's central "working hypotheses."

Also, while the notion that the absence of evidence is not evidence of absence is certainly correct and hence a valid defense (Gartzke and Lupu, 2012: 292, fn. 7), it is a weak defense by any measure. When ACT's proponents keep repeating such an evasive defense, they reduce our confidence in ACT.

The second sign is that ACT's proponents have been trying to repair the enterprise with modified models with more and more restrictive conditions. Unfortunately, the resulting models greatly weaken the original theory or considerably narrow its scope. Slantchev (2006) stressed the necessity of well-balanced free media for audience costs to operate, whereas Tarar and Leventoglu (2009, 2013) underscored that the size of the audience cost is related to the size of the offer and counteroffer. Moreover, these new models have failed to accommodate empirical anomalies presented by ACT's quantitative and qualitative critics.

The third sign is that ACT's supporters have often resorted to the tactics of shifting goalposts without providing a venue for reaching the new posts or changing subjects when facing refuting evidence. Whereas Schultz (2001a) initially continued "looking for audience costs" while warning us of the difficulties of doing so, Schultz (2012) had all but given up the pursuit, noting that "audience costs are not unlike the 'dark matter' of international relations." Similarly, whereas Slantchev (2006) was trying to save audience costs, Slantchev (2012) merely stressed the difficulty of finding audience costs and testing ACT's hypotheses. Likewise, when Gartzke and Lupu (2012: 396) note that "there are several reasons to think domestic audiences may not be as well informed as the theory requires," they are essentially admitting that audience costs may not matter or even exist. Worse, they have failed to grasp that their defense of ACT has again narrowed the scope of ACT considerably and may have even deprived ACT of a key part of its microfoundation. Indeed, Jervis (1970: 74–78) voiced the same doubt against audience costs long ago.

Two tactics of shifting goalposts or changing subjects stand out. First, both Slantchev (2012) and Gartzke and Lupu (2012) contend that audience costs are not a factor but a mechanism (see also Weeks, 2008). Political punishment (i.e. punishing leaders who backed down after escalating, regardless of the regime type) is certainly a core mechanism within ACT.

Yet, a mechanism always needs factors (or variables) to operate, just as a factor always needs a mechanism to drive changes (Bunge 1997). In fact, Fearon (1994b) insists that audience costs are first and foremost a parameter or a factor, and two recent attempts to measure ACT have been explicitly based on the conviction that audience costs are a factor or residue to be measured (Crisman-Cox and Gibilisco, 2018; Kurizaki and Whang, 2015).

Moreover, taking audience costs as a mechanism merely pushes back the most critical question: where is the evidence? ACT's proponents still need to establish the purported effect caused by the mechanism and its enabling or inhibiting factors. If scholars cannot uncover the effects of the supposed mechanism in situations in which it should operate, we are left with serious doubt that the mechanism operates in the real world. Just as a

concept or a variable that cannot be measured is not very useful, a mechanism that cannot be established is not very useful either. Thus, merely insisting that audience costs are a mechanism rather than a variable is at best a weak defense against the criticism that ACT lacks empirical support. Audience costs cannot remain a "dark matter" factor or mechanism.

The second tactic for defending why audience costs cannot be easily established or falsified is the "selection effect" (e.g. Fearon, 2002; Schultz, 2001a). Unfortunately, the selection effect cannot shield ACT from the criticism that audience costs either do not operate or do not matter much (Mercer, 2013: 68–70). Certainly, a theory that tries to hide behind the selection effect cannot be very convincing. At a minimum, ACT's proponents should try to demonstrate that the selection effect has impacted how statesmen behave in their crisis policy deliberations. So far, however, few of ACT's proponents have done so. Instead, ACT's proponents have essentially been claiming that selection effects exist by insisting that the absence of selection effects is exactly the evidence for selection effects![12]

Moreover, as detailed in section two above, selection effects only operate upon leaders in the initial phase of a crisis, that is, whether a challenger decides to challenge and whether the defender escalates the crisis with public threats. After the outcome of a crisis, selection effects no longer operate because now only the public gets to choose whether to reward or punish leaders.

Furthermore, the key thesis advanced by ACT's proponents based on selection effects implies two contradicting or incompatible predictions. On the one hand, according to Fearon (1994b) and Schultz (2001a, 2012), (democratic) leaders' public threats carry more credibility exactly when potential audience costs are high, thus implying that (democratic) leaders are less likely to escalate publicly when they anticipate paying high audience costs if they may have to back down later on. Hence, such cases will be underrepresented in observational studies, according to Schultz (2001a).

On the other hand, ACT also contends that the credibility of one's threat becomes greater precisely in situations in which one's (possible ex post) audience costs are high. This means that (democratic) leaders should escalate more in situations when potential audience costs are high (i.e. tying one's hands with public threats increases the credibility of one's threats, and the other side should be more likely to back down): such a "self-selection (into escalation)" should be rational.

Evidently, these two predictions go against each other. First, audience costs rarely materialize when they are very high because leaders are unwilling to escalate when anticipating audience costs, even though they can bolster credibility significantly. Second, leaders should more frequently escalate publicly because their threats carry more credibility when audience costs are potentially quite high. When this is the case, the principal hypotheses of ACT cannot be tested with observational studies straightforwardly, not only because selection effects make cases of escalating with high audience costs rare but also because ACT's predictions are indeterminate.[13] In fact, this indeterminateness is

12. I owe this particular sentence to Jon Mercer (personal communication, 20 April 2023).
13. This is the key motivation of moving empirical testing of ACT to experimental testing by Tomz (2007). But testing ACT with experimental studies carries its own risks and selection effects, which can be addressed only separately. For earlier useful discussions, see Nomikos and Sambanis (2019) and Hauenstein (2020).

exactly the problem of infinite recursion: should I bluff more or should I bluff less now that you know I have consistently played it straight or that I have bluffed from time to time? There is no determinate answer to the question (Mercer, 2012).

In short, ACT's empirical predictions are indeterminate, partly because they self-contradict. Consider the following: (1) leaders may fail to anticipate audience costs; (2) they choose not to escalate and hence incur audience costs *exactly because they anticipate possible high audience costs*, even though their threats in these situations will be more credible (cf. Schultz, 2012: 373); (3) they can spin out of audience costs when things get tight (Wong, 2024); and (4) different domestic audiences care for different things, from inconsistency to honesty, competency, and belligerency (Nomikos and Sambanis, 2019; Schlesinger and Levy, 2021). With such indeterminate and self-contradicting empirical predictions, it is extremely difficult to pin them down in empirical testing.

Indeed, when addressing the impact of selection effects on detecting audience costs in actual cases, Schultz (2001a) implicitly assumed that leaders will be at least boundedly rational by anticipating the size of audience costs for both sides, or at least for one's own side, mostly correctly. Yet, as Schultz (2001a) implicitly admitted when examining the Fashoda crisis, French leaders had miscalculated the size of audience costs, not only of the British side but also of their own side. Likewise, both Palmerston in the 1863–1864 Schleswig-Holstein crisis and Obama in the 2012–2013 Syria crisis might have made the same error (Borghard and Snyder, 2012; Schlesinger and Levy, 2021). This should not be a surprise: with incomplete information and bounded rationality, and adding cognitive and other biases (Jervis, 1976; Lebow, 2020 [1981b]; Lake, 2010-11; Streich and Levy, 2016), misestimating the size of audience costs should be the norm rather than the exception.

As a result, many of ACT's proponents have not been logically consistent. For instance, Slantchev (2006, 2011: esp. chap. 2) has argued that audience costs are one, but not the only, (important) mechanism (see also Schultz, 2012: 369–372; Walsh, 2013), thus questioning Fearon's (1994b, 1997) implicit thesis that audience costs are a major, if not a central, mechanism for explaining democratic peace and democratic advantages in crisis bargaining.[14] Yet, when responding to Trachtenberg's (2012a) critique of ACT, Slantchev strongly defended ACT.

More strikingly, Schultz (2001a: 34) noted that all indirect statistical evidence (for ACT) is "purely correlational" and advocated for more "direct test[ing]" with historical case studies (Schultz, 2001a: 53).[15] After Snyder and Borghard (2011) and Trachtenberg (2012a) presented their case study evidence against ACT, however, Schultz (2012) now insists that audience costs are a "dark matter" (i.e. hard to observe or establish) yet still a useful idea. Likewise, after admitting that "historical case studies can be better suited to

14. Fearon, (1994a; 1994b) positions that audience costs are critical in crisis bargaining because they are new information, whereas both interest and power have been "priced-in" when a state challenges and its opponent responds with escalation. This is a theoretical bet, which will be addressed in more detail elsewhere (Tang, n.d.a). I thank a reviewer for demanding me to highlight this point here.

15. In fact, Schultz (2001a, 53, fn. 17) perceptively pointed out that a simple correlational study will incorrectly identify the Abadan crisis as a case for supporting ACT, while a case study will provide a more accurate picture.

detecting causal mechanisms than quantitative methods," Gartzke and Lupu (2012) then asserted that "process tracing is much less effective in assessing audience costs than Trachtenberg and others believe (pp. 391–392)." One is at a loss to make sense of their two positions because they are self-contradicting.

Indeed, as summarized in Table 2, some of the cases examined by Schultz (2001a, 2001b), Snyder and Borghard (2011), and Trachtenberg (2012a) are either most likely cases for ACT or provide doubly-decisive tests for or against ACT. In several cases, leaders should have anticipated paying high audience costs if they escalated and then backed down, and yet they still escalated and then backed down (e.g. Attlee in the Abadan crisis, Eden in the Suez crisis; for a comparison of the two crises, see Beck, 2006). Thus, even if some of the cases selected by Snyder and Borghard (2011) and Trachtenberg (2012a) are inappropriate for testing ACT, the overwhelming negative evidence from these two exercises poses serious challenges for ACT (Lorentzen et al., 2017: 476, 483).

Conclusion

For three decades, ACT has been a prominent approach in the study of crisis escalation, conflict, and domestic politics. The validity of ACT is of more than academic interest: it has major implications for how states conduct crisis bargaining, especially between two democracies or between one democracy and one nondemocracy. Yet, empirical evidence for ACT has been inconclusive at best, to put it kindly.

The theoretical lessons from our exercise are thus two-fold. First, we shall be mindful of the possible traps, covered with intuitive insights produced by elegant models. ACT may be one of such traps. As the late Robert Jervis (1988) warned us a while ago, "When a good idea like this comes along, our exuberance at finding new insights leads us to extend and apply it widely, postponing a consideration of problems and limitations" (p. 318). Domestic audience cost is one such appealing idea that may not matter that much in the real world.

Second, most elegant models simplify the real world quite drastically: they are "toys." Doing so may lead to important insights but may also distort reality so much that these insights are just beautiful illusions. A modeling exercise has to strike a balance between elegance (and mathematical tractability) and capturing the real world to a degree.

Despite lacking firm support, however, many authors still routinely deploy ACT as a ready-to-go explanation for crisis bargaining, especially when there is at least one democracy involved in a crisis. Thus, a deeper question can be posed: why so?

While ACT's intuitive plausibility is important, perhaps there are other, more fundamental reasons behind ACT's influence in the field. The first possibility is that ACT seems to readily point to a democratic advantage in crisis bargaining and thus indirectly supports the democratic peace theory (Fearon, 1994b), because audience costs are more possible in democracies, even if we admit there are also audience costs for autocrats in autocracies (e.g. Weeks, 2008; Weiss, 2014). A second possibility is that audience costs seem to replace the nearly immeasurable costs of reputation for resolve or credibility (e.g. Schelling, 1960, 1966), as Fearon (1992) made it clear in his doctoral dissertation. A third possibility is institutional: most adherents of ACT are based in American universities and think-tanks. Due to Fearon's enormous influence and status, not citing Fearon (1994b) will be a glaring sin of omission. These possibilities will be questions for future inquiries.

Methodologically, our exercise provides an innovative way of gauging the empirical validity of ACT (and other theories). By bringing together the case studies testing ACT with an explicit Bayesian logic, our exercise shows that ACT has only marginal validity in the cases examined in the literature. The methodological lessons from our exercise are self-evident.

For ACT itself, while there have been many attempts to test ACT with case studies, many cases might have been "illustrative" or mostly "straw-in-the-wind" tests because they do not rigorously lay out the rules for judging the evidence. To improve future testing with case studies with process tracing, we need to insist that empirical hypotheses are to be derived from models or theories as explicitly as possible so that we can present case studies as "hoop," "smoking gun," or even "doubly decisive" tests.

By synthesizing different case studies scattered in the literature with an explicit Bayesian logic, QMACS provides a new tool for disciplining and synthesizing evidence of case studies regarding other prominent theories with enough case studies that are scattered in the literature. As argued above, issues with "selection effects" may be especially suitable for QMACS. Most prominently, the debate on inequality and democratic transitions can be greatly improved by a QMACS exercise, because Haggard and Kaufmann (2012) have already compiled a qualitative dataset of the cases in the Third Wave of democratization that is readily adaptable for QMACS.

Another issue that is more related to our discussion here and urgently needs a QMACS analysis is extended deterrence. While much has been written on extended deterrence, which is a more complex situation of crisis bargaining (because it involves at least three players), empirical evidence for theories of extended deterrence has been extremely thin (for a review, see Quackenbush, 2011). While earlier quantitative exercises point to some support for the theory (e.g. Danilovic, 2002; Huth, 1988), selection effects make the results from these exercises suspicious. Meanwhile, case study critics of extended deterrence have not been rigorous enough and certainly do not deploy the Bayesian logic (e.g. Lebow and Stein, 1987). Yet, quite a few cases have been examined in detail and thus are readily suitable for performing QMACS. Indeed, some of the cases examined in this paper are cases of extended deterrence and compellence. Other issues that are suitable for QMACS include democratic peace and revolution.

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be elected to this position. In 2024, he was honored as one of the three Distinguished Scholars at the Global IR Section (GIRS) at the ISA Convention in San Francisco, along with Barry Buzan and Cristina Rojas. He has a very broad research interest and has published widely, with five single-authored volumes so far. His most recent books include *The Institutional Foundation of Economic Development* (Princeton University Press, 2022) and On Social Evolution: Phenomenon and Paradigm (Routledge 2020). His earlier book, *The Social Evolution of International Politics* (Oxford University Press, 2013), receives the "Annual Book Award" from the International Studies Association (ISA) for the year of 2015. He is also the author of *A General Theory of Institutional Change* (Routledge, 2011), *A Theory of Security Strategy for Our Time: Defensive Realism* (Palgrave Macmillan, 2010), and many articles in international relations, comparative politics, computational social sciences, evolutionary biology, institutional economics, sociology, and philosophy of the social sciences.